

**DESIGN AND SIMULATE A SMART FLOOR LEVEL MONITORING
SYSTEM USING ULTRASONIC SENSOR.**

ASSIGNMENT – 7

TEAM MEMBERS

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1. PROJECT OVERVIEW

The Smart Flood Level Monitoring System is an IoT-based system developed to continuously monitor water levels and provide early warnings during potential flood conditions. The system uses an HC-SR04 ultrasonic sensor to measure the distance between the sensor and the water surface. The collected data is processed by an ESP8266 NodeMCU, which calculates the water level and determines the current flood condition.

The system displays the water level and status on an OLED display and uses LED indicators to provide a visual indication of the situation. A low-level-trigger buzzer module is activated when the water level reaches the critical threshold, providing an immediate audible warning.

The system classifies the water level into Normal, Warning, and Critical/Flood conditions. This provides a simple and cost-effective solution for continuous monitoring in areas such as rivers, drainage systems, reservoirs, canals, and flood-prone locations. The proposed system can also be further enhanced with Wi-Fi-based remote monitoring and mobile or cloud notifications for real-time disaster management.

MAIN FEATURES

- Continuous real-time water level monitoring using an ultrasonic sensor.
- Accurate measurement of the distance between the sensor and water surface.
- ESP8266 NodeMCU for processing and controlling the system.
- OLED display for showing water level and system status.

- Three-level condition detection: Normal, Warning, and Critical.
- LED indicators for quick visual identification of water conditions.
- Low-level-trigger buzzer for immediate flood alerts.
- Automatic activation of alerts when the water reaches the critical level.
- Low-cost and compact design suitable for flood-prone areas.
- Can be upgraded with Wi-Fi and cloud-based remote monitoring.

2. PROBLEM STATEMENT

Flooding is a serious natural disaster that can occur due to heavy rainfall, overflowing rivers, blocked drainage systems, and rising water levels. Rapid changes in water level can make it difficult to detect flood conditions early, especially when monitoring is performed manually.

Conventional flood monitoring methods may require continuous human observation and may not provide immediate alerts when the water level rises rapidly. This can result in delayed warnings, increasing the risk of damage to people, buildings, roads, agricultural areas, and other infrastructure.

Therefore, there is a need for a simple, reliable, and low-cost automated system that can continuously monitor water levels and provide timely warnings. The proposed Smart Flood Level Monitoring System uses an HC-SR04 ultrasonic sensor to measure the water level without direct contact with the water. An ESP8266 NodeMCU processes the sensor readings and determines whether the water level is normal, at a warning stage, or at a critical flood level.

The system provides real-time information through an OLED display and uses LED indicators and a low-level-trigger buzzer module to provide immediate visual and audible alerts. This approach reduces the dependence on manual monitoring and can support early flood detection and disaster-management activities. The system can also be further developed by adding Wi-Fi, cloud storage, mobile notifications, and remote monitoring capabilities.

3. PROBLEM SOLUTION

The proposed Smart Flood Level Monitoring System provides an automated solution for continuous and real-time water-level monitoring. An HC-SR04 ultrasonic sensor is positioned above the water surface to measure the distance without direct contact with the water. The ESP8266 NodeMCU receives the sensor data, calculates the water level, and compares it with predefined safety thresholds.

When the water level is within the normal range, the system indicates a safe condition using the green LED. When the water level increases to the warning range, the system displays a warning on the OLED and activates the warning indication. When the water level reaches the critical level, the red LED and low-level-trigger buzzer are activated to provide an immediate flood alert.

This automated approach reduces the need for continuous manual observation, provides faster identification of dangerous water levels, and improves early-warning capability. The system can also be expanded with Wi-Fi, cloud monitoring, and mobile notifications to enable remote monitoring and improve disaster-management response.

- Continuously monitors water level using an ultrasonic sensor.

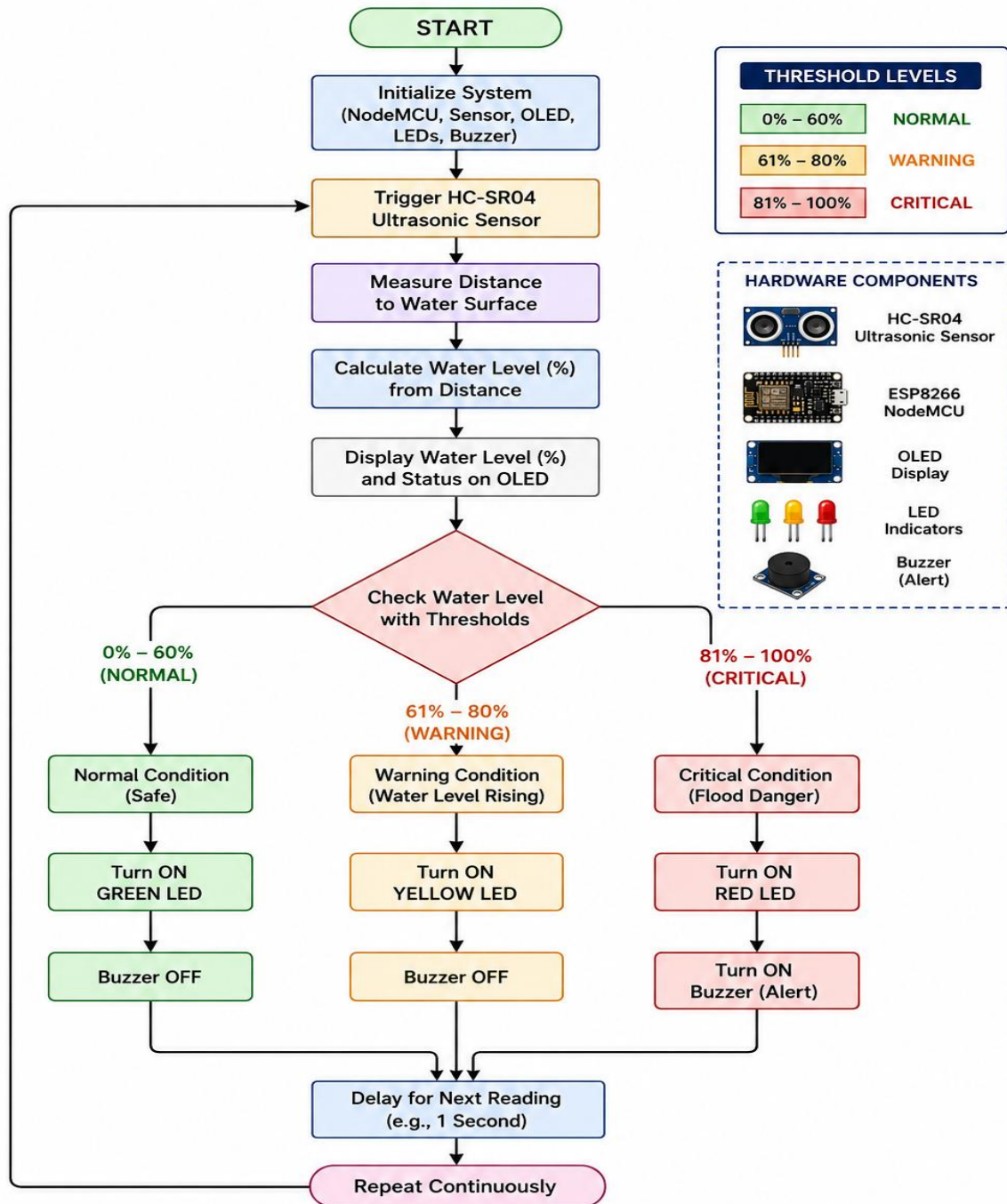
- Measures water level without direct contact with water.
- ESP8266 NodeMCU processes the sensor readings in real time.
- Calculates and displays the water-level percentage on an OLED display.
- Classifies water conditions into Normal, Warning, and Critical levels.
- Provides visual alerts using green and red LED indicators.
- Activates a low-level-trigger buzzer during critical flood conditions.
- Provides early warning to reduce the risk of flood-related damage.
- Reduces the need for continuous manual monitoring.
- Provides a low-cost and compact solution for flood-prone areas.
- Can be integrated with Wi-Fi and cloud platforms for remote monitoring.
- Can be further developed to send mobile notifications during critical conditions.

4. SYSTEM ARCHITECTURE

- HC-SR04 ultrasonic sensor detects the distance between the sensor and the water surface.
- The sensor sends the measured data to the ESP8266 NodeMCU.
- ESP8266 NodeMCU acts as the main processing and control unit.
- The NodeMCU calculates the water level percentage from the measured distance.
- The calculated water level is compared with predefined threshold values.
- OLED display shows the water level and current system status.
- Green LED indicates the normal water-level condition.
- Yellow/Red LED indicates warning or critical water-level conditions, according to the implemented circuit.
- Low-level-trigger buzzer provides an audible alert when the water level reaches the critical condition.
- The system continuously repeats the sensing, processing, display, and alert process for real-time flood monitoring.

FLOW CHART

SMART FLOOD LEVEL MONITORING SYSTEM



5. CIRCUIT TABLE

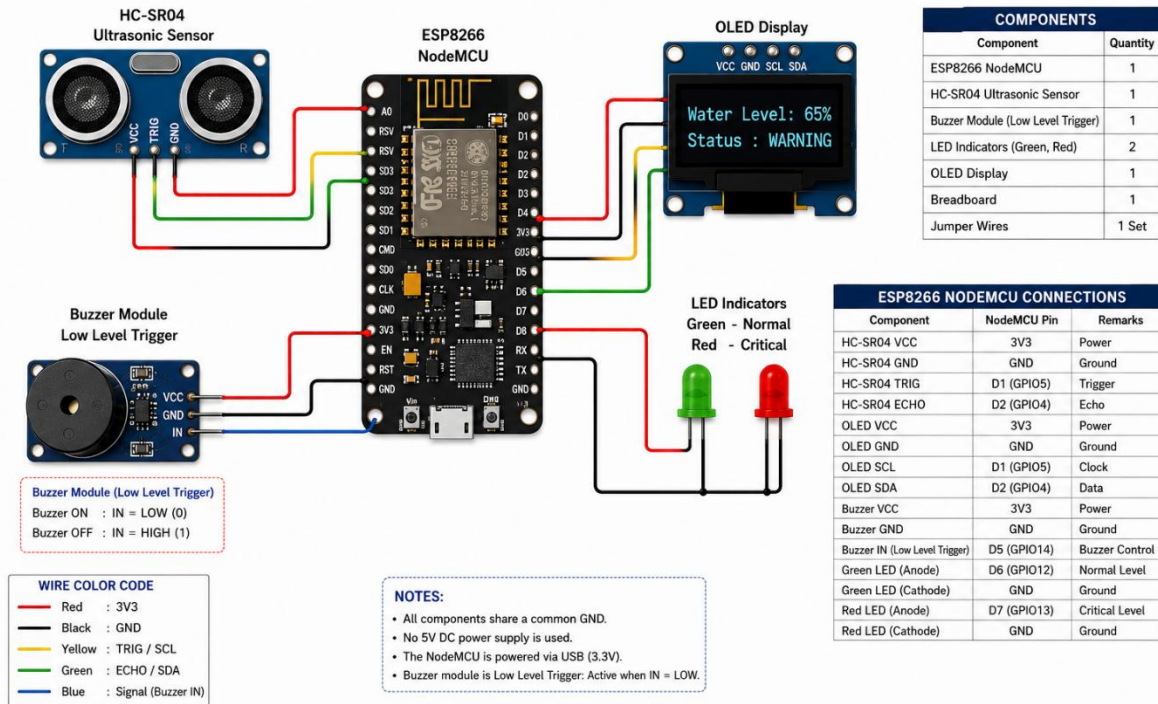
Component	Pin/Terminal	ESP8266 NodeMCU Pin	Purpose
HC-SR04	VCC	3V3	Power
HC-SR04	GND	GND	Ground
HC-SR04	TRIG	D5 (GPIO14)	Trigger signal
HC-SR04	ECHO	D6 (GPIO12)	Echo signal
OLED	VCC	3V3	Power
OLED	GND	GND	Ground
OLED	SCL	D1 (GPIO5)	I ² C Clock
OLED	SDA	D2 (GPIO4)	I ² C Data
Buzzer Module	VCC	3V3	Power
Buzzer Module	GND	GND	Ground
Buzzer Module	IN	D7 (GPIO13)	Buzzer control
Green LED	Anode	D0 (GPIO16)	Normal indication
Green LED	Cathode	GND	Ground
Red LED	Anode	D4 (GPIO2)	Critical indication
Red LED	Cathode	GND	Ground

- The HC-SR04 sensor sends ultrasonic waves toward the water surface.
- The waves reflect back from the water surface and are received by the sensor.
- The ESP8266 NodeMCU calculates the distance using the echo time.
- The measured distance is converted into the water-level percentage.
- The NodeMCU compares the water level with predefined threshold values.
- The OLED display shows the water level and current status.
- The green LED indicates a normal or safe water level.
- The warning indication is activated when the water level rises above the safe limit.
- The red LED indicates a critical water level.
- The low-level-trigger buzzer turns ON when the water level reaches the critical threshold.
- The buzzer provides an immediate audible warning during flood conditions.
- The system continuously repeats the measurement and monitoring process.
- This enables real-time water-level monitoring and early flood detection.

6. CIRCUIT DESIGN

SMART FLOOD LEVEL MONITORING SYSTEM

CIRCUIT CONNECTION DIAGRAM (WITHOUT 5V DC POWER SUPPLY)

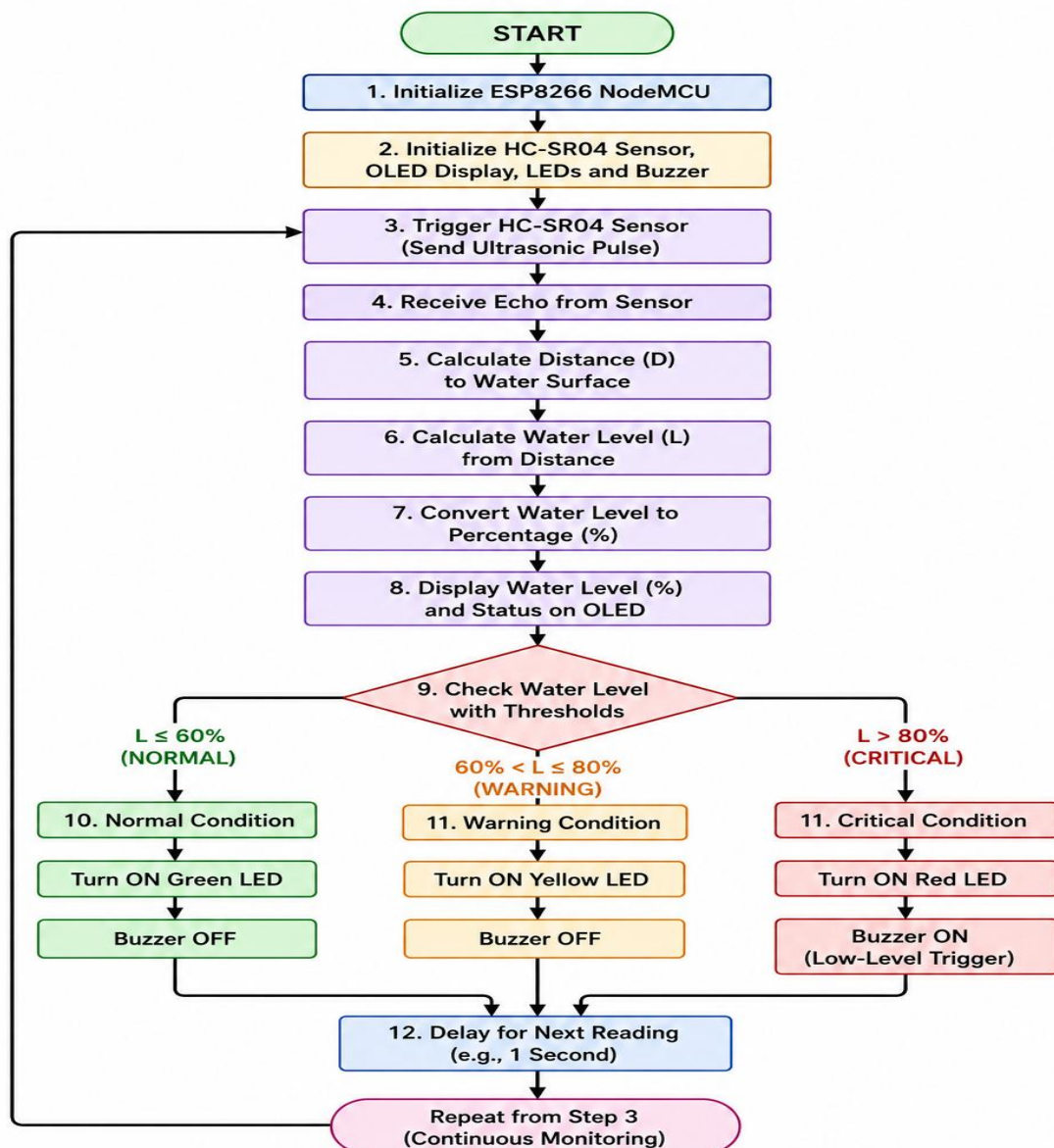


- The ESP8266 NodeMCU is used as the main controller of the system.
- The HC-SR04 ultrasonic sensor is connected to the NodeMCU for water-level measurement.
- The TRIG pin of the ultrasonic sensor is connected to D5, and the ECHO pin is connected to D6.
- The OLED display is connected through I²C communication using D1 (SCL) and D2 (SDA).
- A low-level-trigger buzzer module is connected to D7 for critical-level alerts.
- The green LED connected to D0 indicates the normal water-level condition.
- The red LED connected to D4 indicates the critical water-level condition.
- All components share a common GND with the NodeMCU.
- The circuit is powered through the NodeMCU USB connection, without a separate 5V DC supply.
- The NodeMCU continuously receives sensor readings and controls the display, LEDs, and buzzer according to the water level.
- The circuit provides real-time monitoring and automatic flood-level warning.

7. ALGORITHM

The algorithm begins by initializing the ESP8266 NodeMCU along with the HC-SR04 ultrasonic sensor, OLED display, LED indicators, and low-level-trigger buzzer module. The ultrasonic sensor sends a pulse toward the water surface and receives the reflected echo. The NodeMCU uses the echo time to calculate the distance between the sensor and the water surface, and this distance is converted into the corresponding water-level percentage. The water level and status are then displayed on the OLED screen. The NodeMCU compares the measured level with the predefined threshold values. If the water level is normal, the green LED remains ON and the buzzer remains OFF. If the level reaches the warning range, the warning indication is activated. When the water level reaches the critical range, the red LED is turned ON and the low-level-trigger buzzer is activated to provide an immediate flood alert. After a short delay, the system repeats the entire process continuously for real-time flood monitoring.

ALGORITHM FLOW CHART (SMART FLOOD LEVEL MONITORING SYSTEM)



8. CODING


```
#include <Wire.h>

#include <Adafruit_GFX.h>

#include <Adafruit_SSD1306.h>


// ----- OLED -----

#define SCREEN_WIDTH 128

#define SCREEN_HEIGHT 64


Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);


// ----- HC-SR04 -----

#define TRIG_PIN D5

#define ECHO_PIN D6


// ----- Buzzer Module -----

// LOW LEVEL TRIGGER

#define BUZZER_PIN D7


// ----- LEDs -----

#define GREEN_LED D0

#define RED_LED D4


// ----- Flood Settings -----

// Height from sensor to bottom of monitoring area

const float MAX_HEIGHT_CM = 100.0;
```

```
// Flood warning levels

const float WARNING_LEVEL = 50.0;

const float CRITICAL_LEVEL = 80.0;


// -----

void setup()

{

    Serial.begin(115200);


    // Ultrasonic sensor

    pinMode(TRIG_PIN, OUTPUT);

    pinMode(ECHO_PIN, INPUT);


    // Buzzer

    pinMode(BUZZER_PIN, OUTPUT);


    // LEDs

    pinMode(GREEN_LED, OUTPUT);

    pinMode(RED_LED, OUTPUT);


    // Initial states

    digitalWrite(TRIG_PIN, LOW);
```

```
// Low-level trigger:

// HIGH = OFF

digitalWrite(BUZZER_PIN, HIGH);


digitalWrite(GREEN_LED, LOW);
digitalWrite(RED_LED, LOW);


// OLED I2C
Wire.begin(D2, D1);

if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
{
    Serial.println("OLED not found!");
    while (1);
}

display.clearDisplay();
display.setTextColor(SSD1306_WHITE);


display.setTextSize(1);
display.setCursor(20, 20);
display.println("SMART FLOOD");
display.setCursor(15, 35);
display.println("MONITORING");
```

```
display.display();
```

```
delay(2000);
```

```
}
```

```
// -----
```

```
float getDistance()
```

```
{
```

```
    // Send ultrasonic pulse
```

```
    digitalWrite(TRIG_PIN, LOW);
```

```
    delayMicroseconds(2);
```

```
    digitalWrite(TRIG_PIN, HIGH);
```

```
    delayMicroseconds(10);
```

```
    digitalWrite(TRIG_PIN, LOW);
```

```
    // Read echo
```

```
    long duration = pulseIn(ECHO_PIN, HIGH, 30000);
```

```
    // If no echo received
```

```
    if (duration == 0)
```

```
    {
```

```
        return -1;
```

```
}
```

```
// Distance in cm
```

```
float distance = duration * 0.0343 / 2.0;
```

```
return distance;
```

```
}
```

```
// -----
```

```
void loop()
```

```
{
```

```
float distance = getDistance();
```

```
// Sensor error
```

```
if (distance < 0)
```

```
{
```

```
Serial.println("Ultrasonic Sensor Error");
```

```
display.clearDisplay();
```

```
display.setTextSize(1);
```

```
display.setCursor(15, 20);
```

```
display.println("SENSOR ERROR!");
```

```
display.setCursor(10, 40);
```

```
display.println("Check HC-SR04");
```

```
display.display();
```

```
digitalWrite(GREEN_LED, LOW);
```

```
digitalWrite(RED_LED, LOW);
```

```
// Buzzer OFF
```

```
digitalWrite(BUZZER_PIN, HIGH);
```

```
delay(1000);
```

```
return;
```

```
}
```

```
// Limit distance
```

```
if (distance > MAX_HEIGHT_CM)
```

```
{
```

```
    distance = MAX_HEIGHT_CM;
```

```
}
```

```
// Calculate water level
```

```
float waterLevel = MAX_HEIGHT_CM - distance;
```

```
// Calculate percentage
```

```
float percentage = (waterLevel / MAX_HEIGHT_CM) * 100.0;
```

```
// -----  
  
// NORMAL CONDITION  
  
// -----  
  
if (percentage < WARNING_LEVEL)  
{  
    digitalWrite(GREEN_LED, HIGH);  
    digitalWrite(RED_LED, LOW);  
  
    // Low-level trigger buzzer OFF  
    digitalWrite(BUZZER_PIN, HIGH);  
  
    display.clearDisplay();  
  
    display.setTextSize(1);  
    display.setCursor(0, 0);  
    display.println("SMART FLOOD MONITOR");  
  
    display.setTextSize(2);  
    display.setCursor(0, 18);  
    display.print("Level:");  
    display.print((int)percentage);  
    display.println("%");  
  
    display.setTextSize(1);
```

```
display.setCursor(0, 45);

display.println("STATUS: NORMAL");


display.display();


Serial.print("Water Level: ");
Serial.print(percentage);
Serial.println("% - NORMAL");
}


// -----
// WARNING CONDITION
// -----


else if (percentage < CRITICAL_LEVEL)
{
    digitalWrite(GREEN_LED, LOW);
    digitalWrite(RED_LED, HIGH);


    // Buzzer OFF during warning
    digitalWrite(BUZZER_PIN, HIGH);


    display.clearDisplay();


    display.setTextSize(1);
```



```
display.setCursor(0, 0);  
display.println("SMART FLOOD MONITOR");
```

```
display.setTextSize(2);  
display.setCursor(0, 18);  
display.print("Level:");  
display.print((int)percentage);  
display.println("%");
```

```
display.setTextSize(1);  
display.setCursor(0, 45);  
display.println("STATUS: WARNING");
```

```
display.display();
```

```
Serial.print("Water Level: ");  
Serial.print(percentage);  
Serial.println("% - WARNING");
```

```
}
```

```
// -----
```

```
// CRITICAL / FLOOD CONDITION
```

```
// -----
```

```
else
```

```
{  
  
    digitalWrite(GREEN_LED, LOW);  
  
    digitalWrite(RED_LED, HIGH);  
  
  
    // LOW LEVEL TRIGGER  
  
    // LOW = BUZZER ON  
  
    digitalWrite(BUZZER_PIN, LOW);  
  
  
    display.clearDisplay();  
  
  
    display.setTextSize(1);  
    display.setCursor(0, 0);  
    display.println("!!! FLOOD ALERT !!!");  
  
  
    display.setTextSize(2);  
    display.setCursor(0, 18);  
    display.print("Level:");  
    display.print((int)percentage);  
    display.println("%");  
  
  
    display.setTextSize(1);  
    display.setCursor(0, 45);  
    display.println("CRITICAL WATER LEVEL");  
  
  
    display.display();
```

```

Serial.print("Water Level: ");

Serial.print(percentage);

Serial.println("% - FLOOD ALERT!");

}

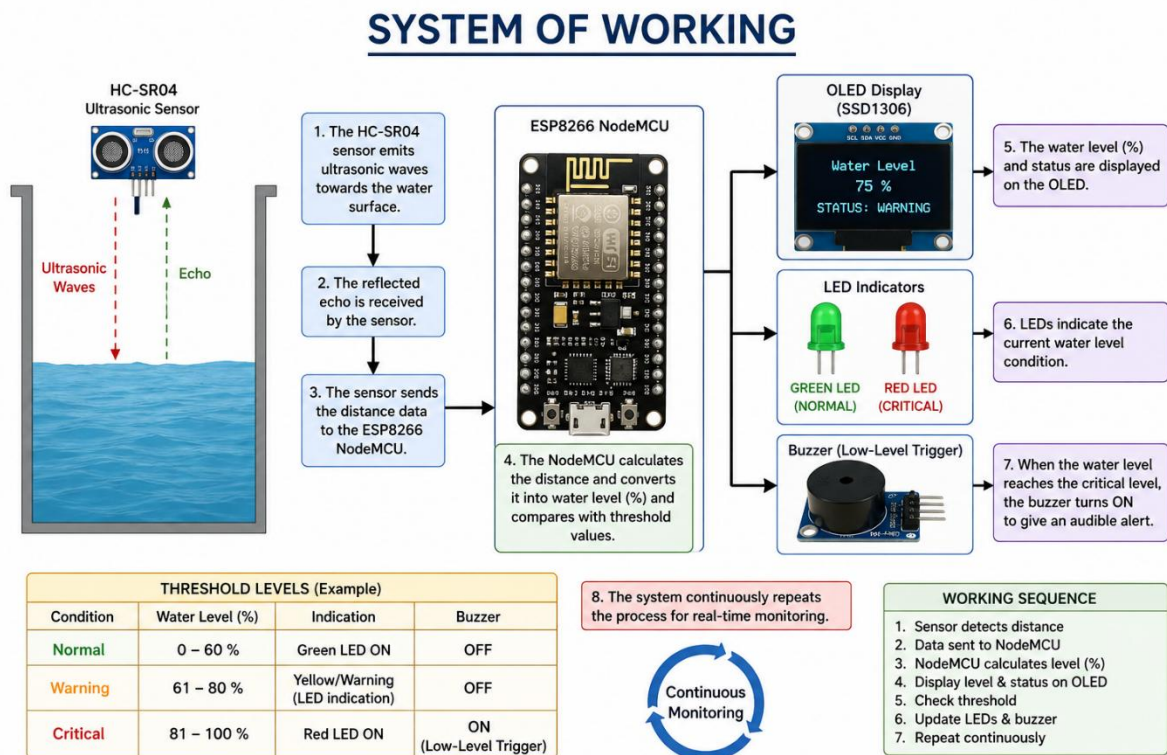
// Update every 1 second

delay(1000);

}

```

9. SYSTEM WORKING



The Smart Flood Level Monitoring System works by continuously detecting and monitoring the water level using an HC-SR04 ultrasonic sensor. The sensor is placed above the water surface and sends ultrasonic waves toward the water. When the waves reflect back, the sensor provides the echo signal to the ESP8266 NodeMCU. The NodeMCU calculates the distance between the sensor and water surface and converts it into the corresponding water-level percentage.

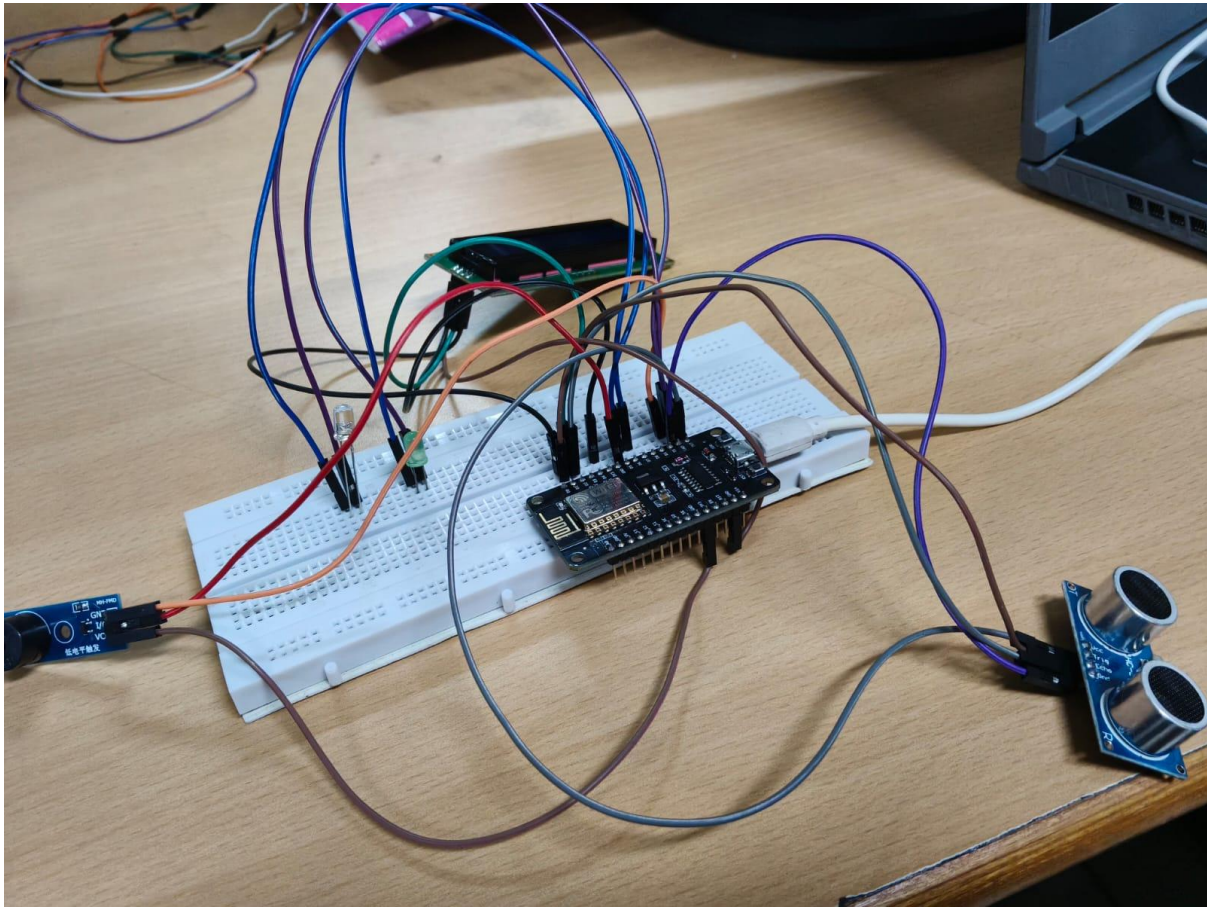
The calculated water level is compared with predefined threshold values to determine the current condition. The OLED display shows the water-level percentage and system status in real time. During normal conditions, the green LED indicates a safe water level and the buzzer remains OFF. When the water level increases to the warning range, the system provides a warning indication. When the water level reaches the critical level, the red LED is activated and the low-level-trigger buzzer turns ON to provide an audible flood alert.

The system repeats this sensing, processing, displaying, and alerting process continuously, allowing real-time monitoring and early detection of potential flood conditions.

10. BILL OF MATERIALS

COMPONENTS	QUANTITY
ESP8266 NodeMCU	1
HC-SR04 Ultrasonic Sensor	1
Buzzer	1
LED Indicators	2
OLED/LCD Display	1
Breadboard	1
Jumper Wires	1 SET

11. PROTOTYPE IMPLEMENTATION

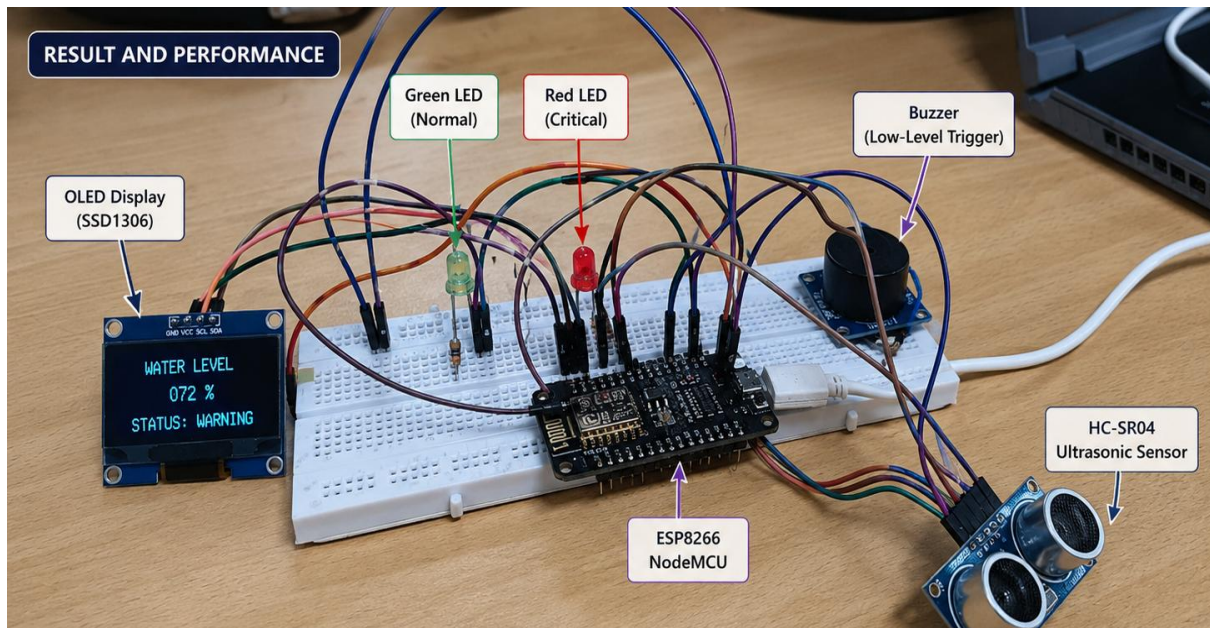


The prototype of the Smart Flood Level Monitoring System is implemented using an ESP8266 NodeMCU, HC-SR04 ultrasonic sensor, OLED display, LED indicators, buzzer module, breadboard, and jumper wires. The HC-SR04 sensor is positioned above the water container to continuously measure the distance between the sensor and the water surface.

The ESP8266 NodeMCU receives the sensor readings and calculates the corresponding water level. The measured value is compared with predefined threshold levels to identify normal, warning, and critical conditions. The OLED display provides real-time information about the water level and system status.

During normal conditions, the green LED indicates a safe water level. As the water level rises, the warning indication is activated. When the water reaches the critical level, the red LED and low-level-trigger buzzer are activated to provide an immediate flood warning. The prototype continuously repeats the measurement and monitoring process, demonstrating the practical operation of the proposed flood-level monitoring system.

12. RESULT & PERFORMANCE



The prototype successfully monitored the water level using the HC-SR04 ultrasonic sensor and ESP8266 NodeMCU. The sensor continuously measured the distance to the water surface, and the NodeMCU converted the readings into the corresponding water-level percentage.

- The system successfully detected changes in water level in real time.
- The OLED display showed the measured water level and current status.
- The green LED indicated the normal water-level condition.
- The warning indication was activated when the water level increased beyond the safe range.
- The red LED indicated the critical water-level condition.
- The low-level-trigger buzzer successfully provided an audible alert during critical conditions.
- The ESP8266 NodeMCU processed the sensor readings continuously without manual intervention.
- The prototype demonstrated reliable automatic flood-level detection under controlled testing.
- The system provided faster warning compared with manual water-level observation.
- Overall, the prototype demonstrated that the proposed system can be used as a low-cost solution for real-time flood-level monitoring and early warning.